

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

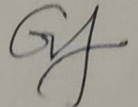
Annexure A

Scenario-Based

Code of Conduct:

I P. G. Sreeshma (192424418) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure infrastructure for its academic and administrative departments. The administrator has been allocated the IP address block 192.168.100.0/24 on the expected number of devices, the following subnet masks have been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.

3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not have any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

		context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Question 1:-

Given

- IP Address = 192.168.10.65
- Subnet Mask = 255.255.255.192 (/26)
- Default Gateway = 192.168.10.1

Identify the subnet:-

A /26 subnet has blocks of 64 Addresses

subnets:-

- 192.168.10.0/26
- 192.168.10.64/26
- 192.168.10.128/26
- 192.168.10.192/26

Since 65 lies between 64 and 127 the Workstation belongs to
192.168.10.64/26

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Valid Host Range

192.168.10.65 - 192.168.10.126

Verify IP and Gateway.

IP Address - 192.168.10.65 → Valid

Gateway - 192.168.10.1 → Invalid.

Configuration Error

The default gateway is configured is a different Subnet

Usable Hosts

$$2^6 - 2 = 64 - 2 = 62, \text{ Usable Hosts} = 62$$

Question - 2

Device - A : 2001:db8::ff00:42:8329/64

Device - B : 2001:db8:0:0:1::1/64

Expanded : 2001:0db8:0000:0000:0000:0000:ff00:0042:8329

Expanded :- 2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix:-

Device A : 2001:db8:0000:0000::/64

Device B : 2001:db8:0000:0000::/64

Same Subnet:-

Yes, Both share the same first 64 bits

Addressing Error:-

No subnet Mismatch.
Communication failure may be due to.

→ wrong gateway

→ Firewall

Optimized Address:-

Device A : 2001:db8::10/64

Device B : 2001:db8::20/64

Number of Hosts

$$2^{64} = 18,446,744,073,709,551,616$$

Question 3:-

Network : 192.168.1.0/24

Subnets : • /26

• /27

• /28

1/26

Network : 192.168.1.0

Broadcast : 192.168.1.63

Host Range : 192.168.1.1 - 192.168.1.62

Usable Hosts : 62

unused : Depends on department size.

/27

Network : 192.168.1.64

Broadcast : 192.168.1.95

Host Range : 192.168.1.65 - 192.168.1.94

Usable Hosts : 30.

/28

Network : 192.168.1.96

Broadcast : 192.168.1.111

Host Range :- 192.168.1.97 - 192.168.1.110

Usable Hosts :- 14.

Inefficient :-

- Small departments receiving /26 waste addresses
- Large departments using /28 experience IP shortage

Question - 4

Expanded Address :-

Device A : 2001 : 0db8 : 0000 : 0000 : 0000 : ff00 : 0042 : 8329

Device B : 2001 : 0db8 : 0000 : 0000 : 0001 : 0000 : 0000 : 0001

Network prefix : 2001 : db8 : 0000 : 0000 :: /64

Same subnet : yes

Possible Error :-

- Interface disabled
- Firewall

Optimized Addresses:-

Device A : 2001::db8::10/64

Device B : 2001::db8::20/64

Host capacity

$2^{64} = 18,446,744,073,709,551,616$ Hosts.

Question 5

Routing Table :-

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Longest prefix match :-

Matching network 192.168.2.0/24 Next Hop = 10.0.0.3

Verify Destination :-

192.168.2.100 belongs to

192.168.2.0/24

Therefore it is a valid destination

Routing Error :-

If packets do not reach the destination possible cause.

- incorrect next-hop configuration
- interface down

Default Route :-

If no routing entry matches, the router forwards packets using

$0.0.0.0/0 \rightarrow 10.0.0.1$